

Hyperparameter Tuning Experiments - Ntwari Mike Chris Kevin

Experiments conducted:

<https://drive.google.com/drive/folders/1-Qdx-QghN4YOLNzGWWe7l4grgAxxE-pt?usp=sharing>

Exp	Policy	lr	Gamma	Batch Size	Epsilon Start	Epsilon End	Epsilon Decay	Best Eval Reward	Noted Observation
1	CnnPolicy	0.0001	0.99	32	1	0.05	0.1	-13.2	The model used a moderate learning rate and standard exploration settings. Training was stable, but performance was limited compared to later experiments. The agent learned some useful behaviors, but the reward remained relatively low.
2	CnnPolicy	0.001	0.95	32	1	0.05	0.1	-21	The higher learning rate (0.001) caused unstable learning. Although the discount factor was reduced to 0.95, the agent struggled to improve performance, resulting in a lower reward.
3	CnnPolicy	0.0005	0.99	32	1	0.05	0.1	-8.8	Reducing the learning rate to 0.0005 significantly improved performance. The model achieved better reward compared to Experiment 1 and 2, showing that a smaller learning rate provided more stable updates while maintaining effective learning.
4	MlpPolicy	0.0005	0.99	32	1	0.05	0.1	-20.8	The MLP policy performed poorly compared to the CNN policy with the same learning rate and gamma. This indicates that convolutional layers are more suitable for processing Atari visual inputs because they can extract spatial features from game frames.
5	CnnPolicy	0.001	0.99	32	1	0.05	0.1	-21	Increasing the learning rate back to 0.001 while keeping gamma at 0.99 reduced performance. The result confirms that a high learning rate caused instability and prevented the agent from effectively learning the game strategy.

6	CnnPolicy	0.0005	0.99	32	1	0.1	0.1	-10.8	Increasing the final epsilon from 0.05 to 0.10 improved exploration and slightly increased performance. The agent maintained more exploration during training, allowing it to discover better actions compared to Experiment 3.
7	CnnPolicy	0.0005	0.99	64	1	0.05	0.1	-3.8	Increasing the batch size from 32 to 64 produced the best performance so far. The larger batch provided more stable gradient updates and reduced noise during learning, resulting in a significant improvement in evaluation reward.
8	CnnPolicy	0.0005	0.995	64	1	0.1	0.1	-5	Combining a larger batch size, higher gamma (0.995), and increased exploration further improved performance. The higher discount factor helped the agent consider future rewards, while increased exploration helped discover better strategies.
9	CnnPolicy	0.001	0.995	64	1	0.1	0.1	-21	Using a higher learning rate (0.001) with gamma 0.995 negatively affected performance. Despite using a larger batch size and more exploration, the aggressive learning rate caused unstable training and poor reward results.
10	MlpPolicy	0.0005	0.995	64	1	0.1	0.1	-21	The MLP policy with the same hyperparameters as Experiment 8 performed poorly compared to CNN. This further confirmed that CNN policies are better suited for Atari environments because they process image observations more effectively.